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Artificial Intelligence and Machine Learning

What is it? What does it do? What are the risks?

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Associate Professor, Computer Engineering and Computer Science

October 30, 2023

What is artificial intelligence (AI)?

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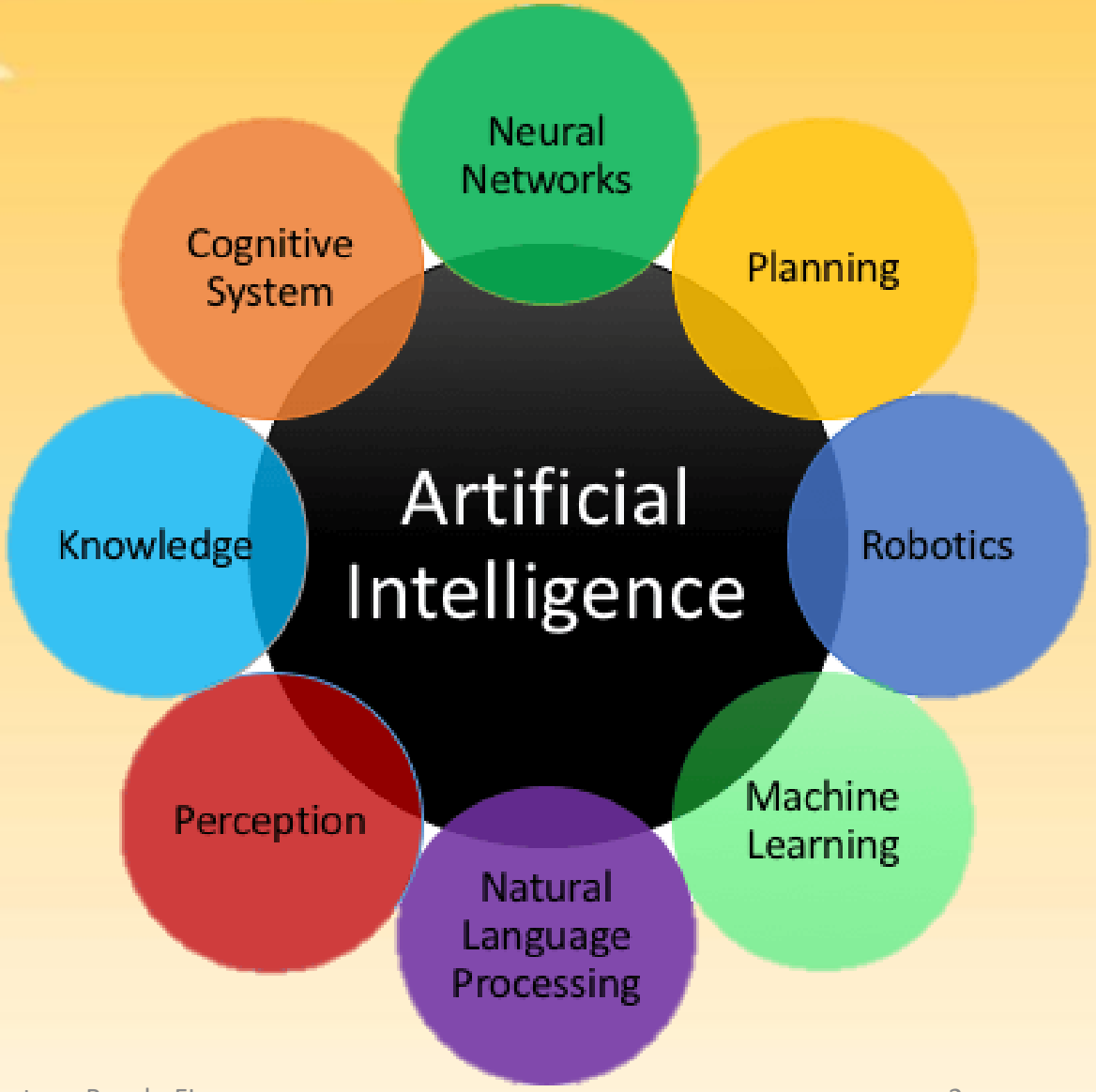
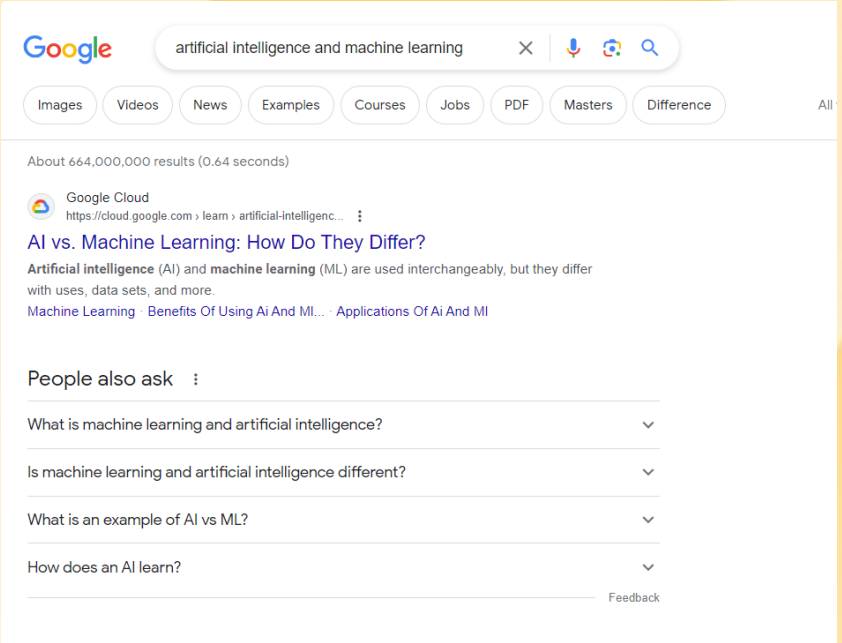
Machines that mimic human intelligence

- Merriam-Webster: Intelligence
  - *(1) the ability to **learn** or **understand** or to deal with new or trying situations : reason; also : the skilled use of reason*
  - *(2) the ability to **apply knowledge** to manipulate one's environment or to think abstractly as measured by objective criteria (as tests)*

- **Weak AI** systems can solve specific problems that would require human intelligence or greater to solve.
- ***Artificial General Intelligence*** systems would be capable of intelligently solving all problems that humans can solve.



Everyday AI



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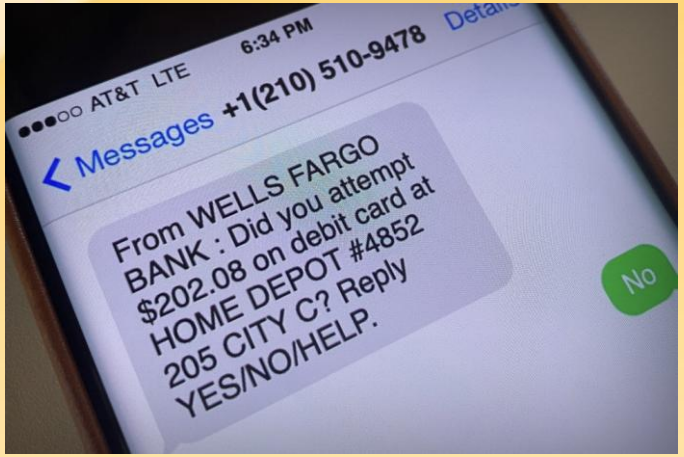
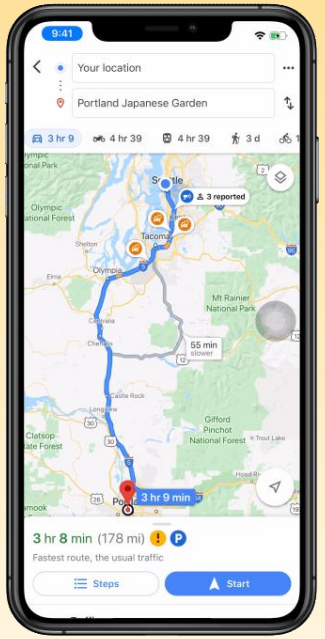
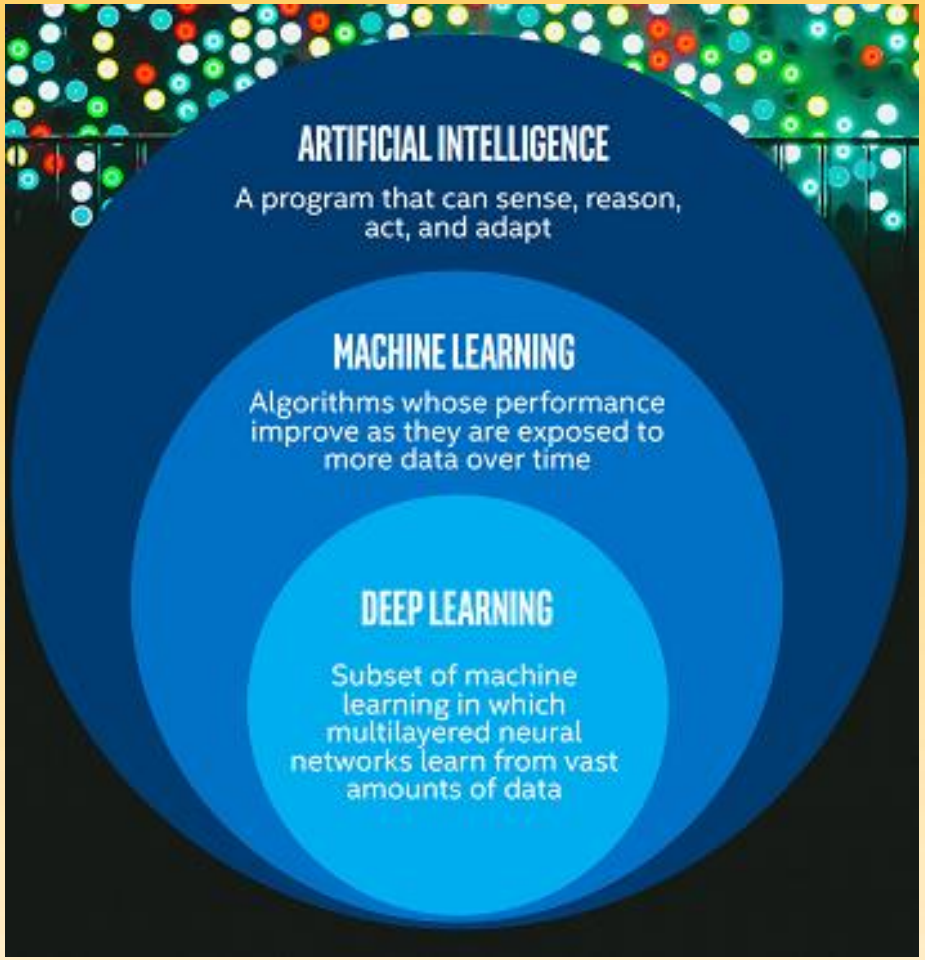




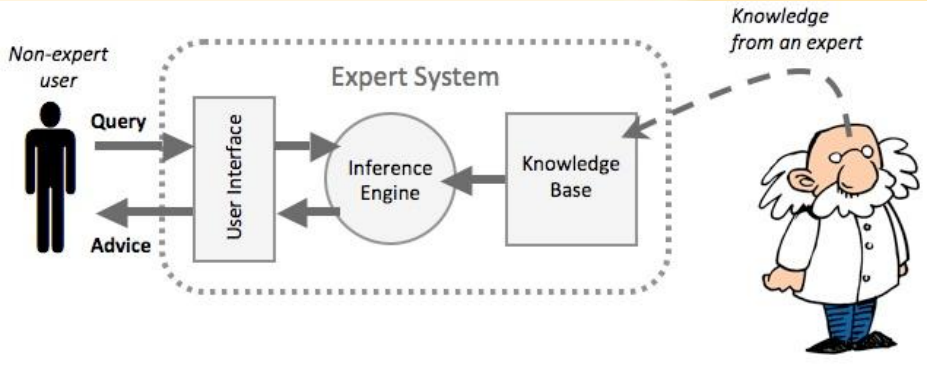
Machine Learning Defined

- Webster-Miriam:
  - "A computational method that is a subfield of artificial intelligence and that enables a computer to learn to perform tasks by analyzing a large dataset without being explicitly programmed.
- Using historically collected data or past states of a system, an ML model is trained to solve a problem.
- Three major types:
  - Supervised Learning.
  - Unsupervised Learning
  - Reinforcement Learning



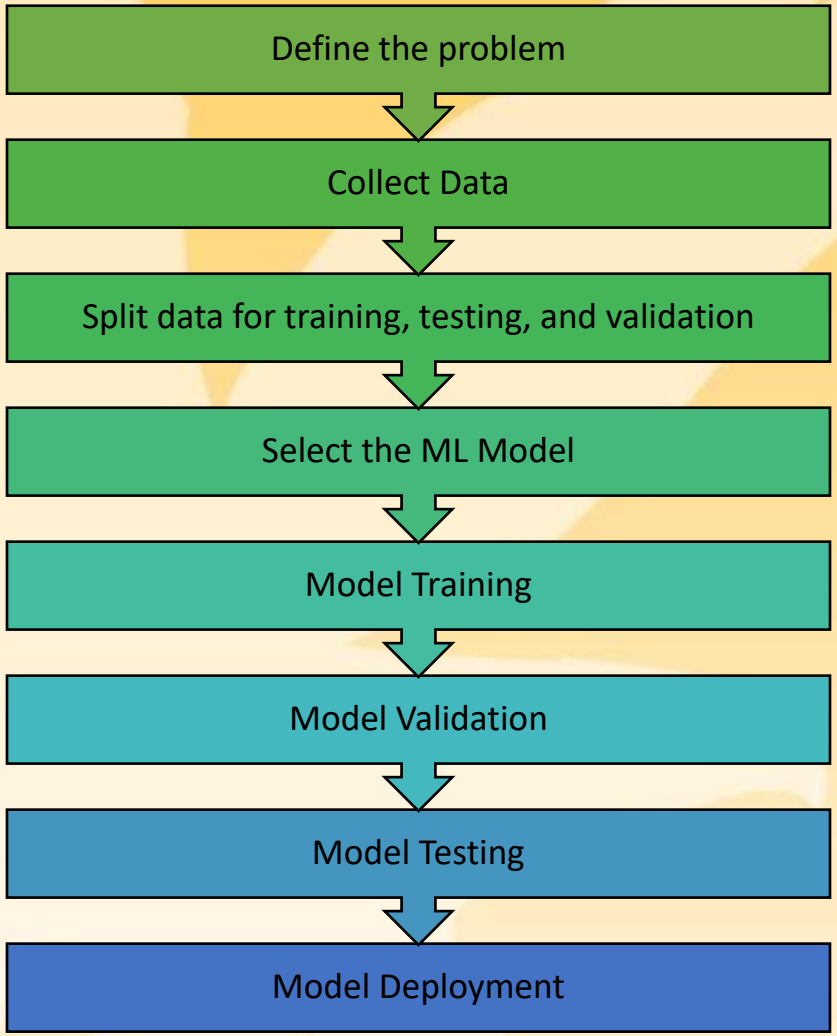
The Power of Data

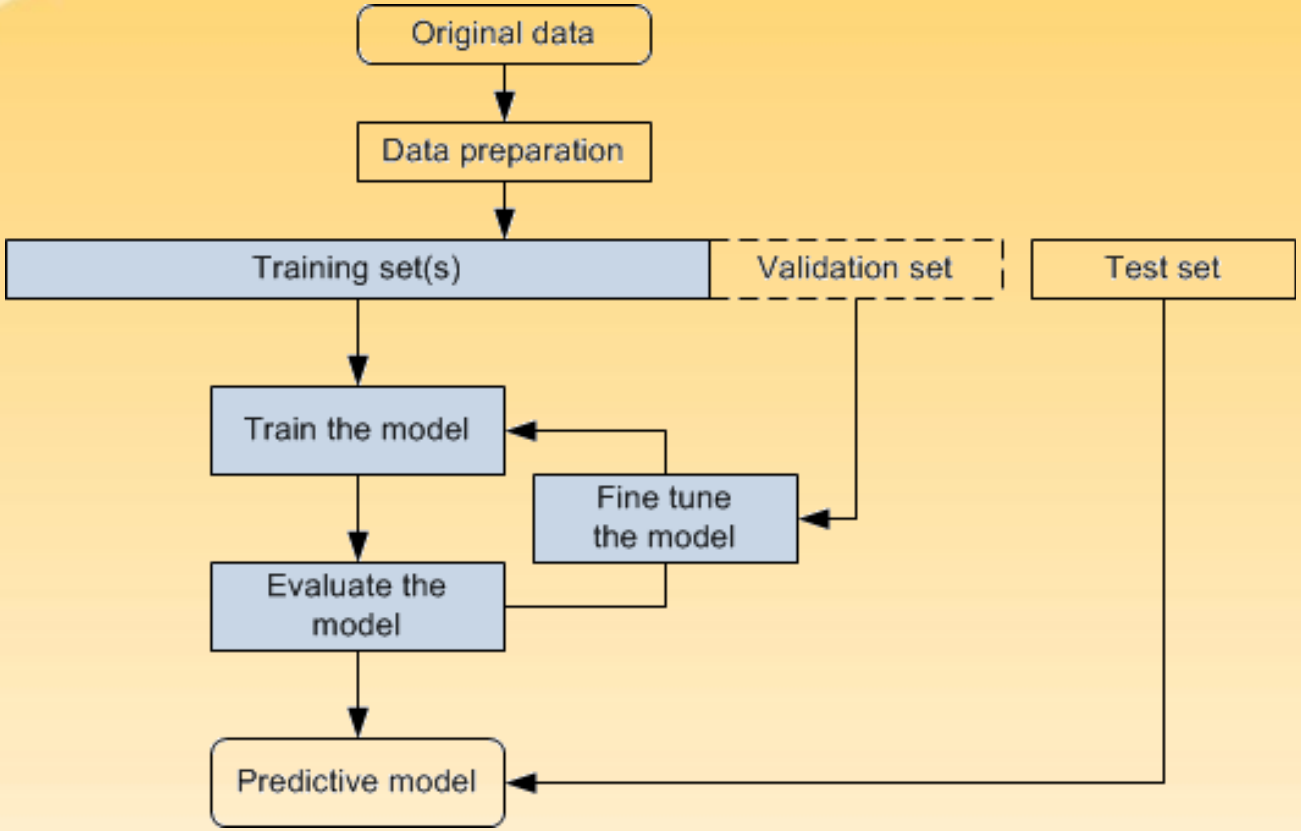
- Machine learning models largely are data-driven.
  - Insufficient data for a model can result in overfitting to only the cases given to it.
  - Data needs scale with model complexity
- Exploration of data should be done with a plan and purpose in mind.
- Across many industries, data is a commodity.



- Challenge: Garbage In→Garbage Out
  - Missing data
  - Imbalanced Data
  - Outliers
  - Noisy data
  - Scaling
  - Redundancy
  - Lack of Diversity
  - Poor data labeling
  - Ethical and legal considerations

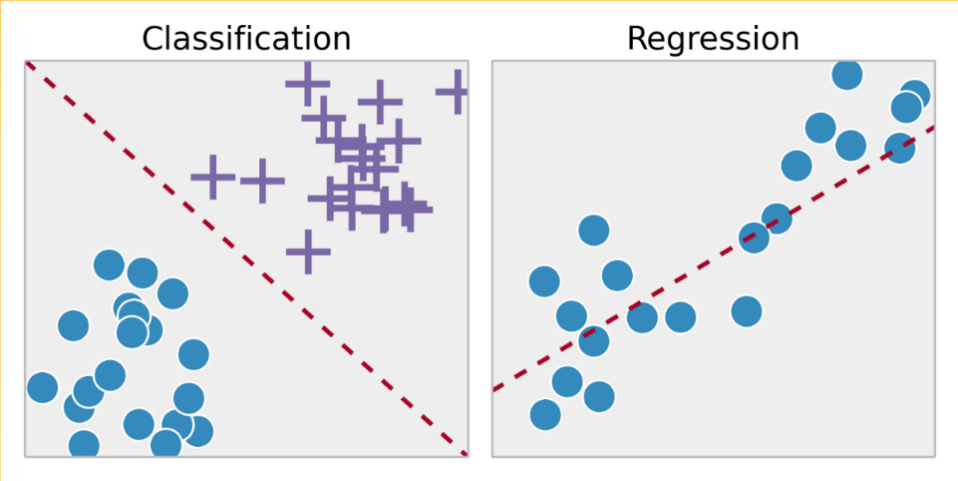
How Machine Learning Works

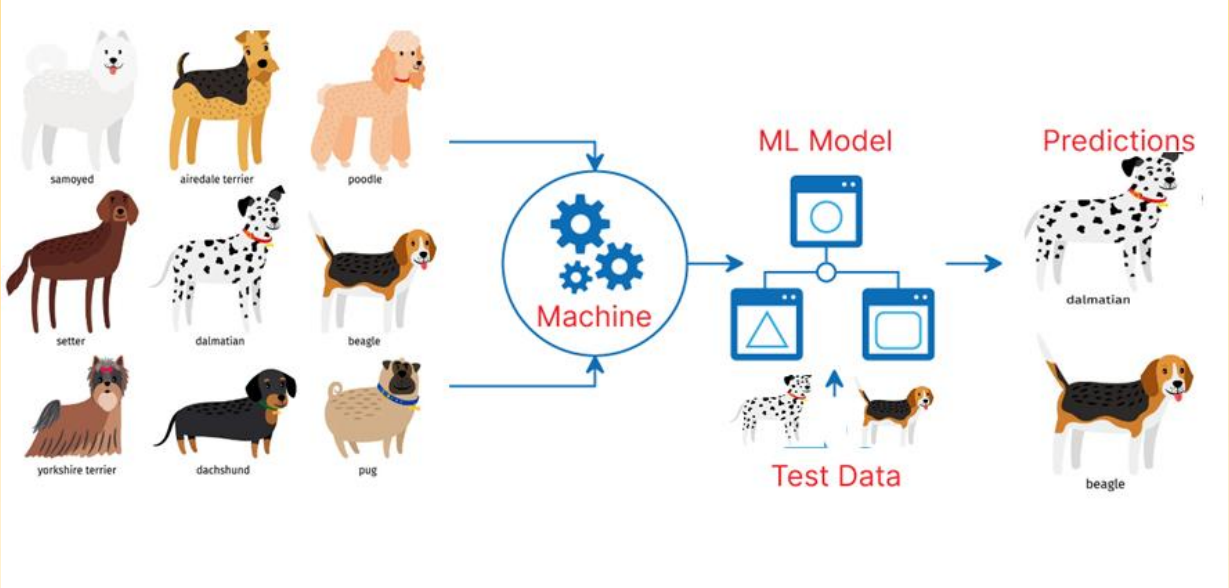




Supervised Learning

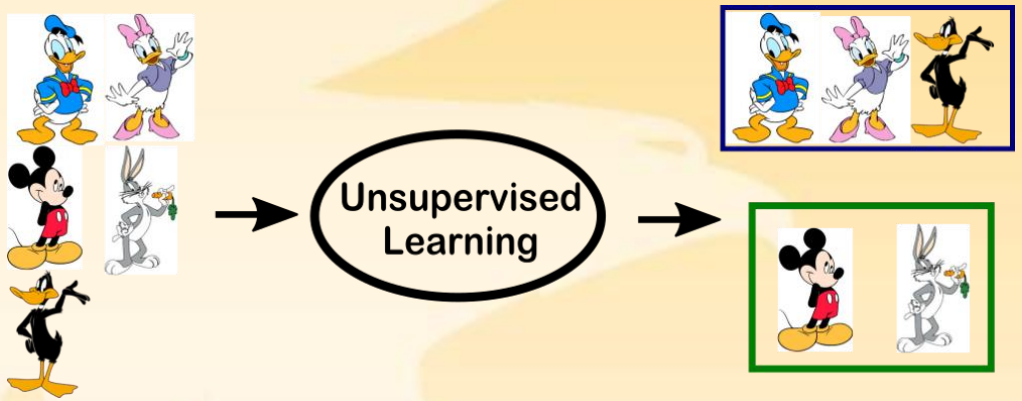
- **Training using labeled data (inputs and expected outputs) to**
  - Make a prediction
  - Determine the measure of error or loss
  - Make corrections based upon the errors
- ***Classification*** – determines the category of a given input (e.g. classify a radar target by aircraft type)
- ***Regression*** – fits the model to the data to produce a continuous value (e.g. predicting house prices)

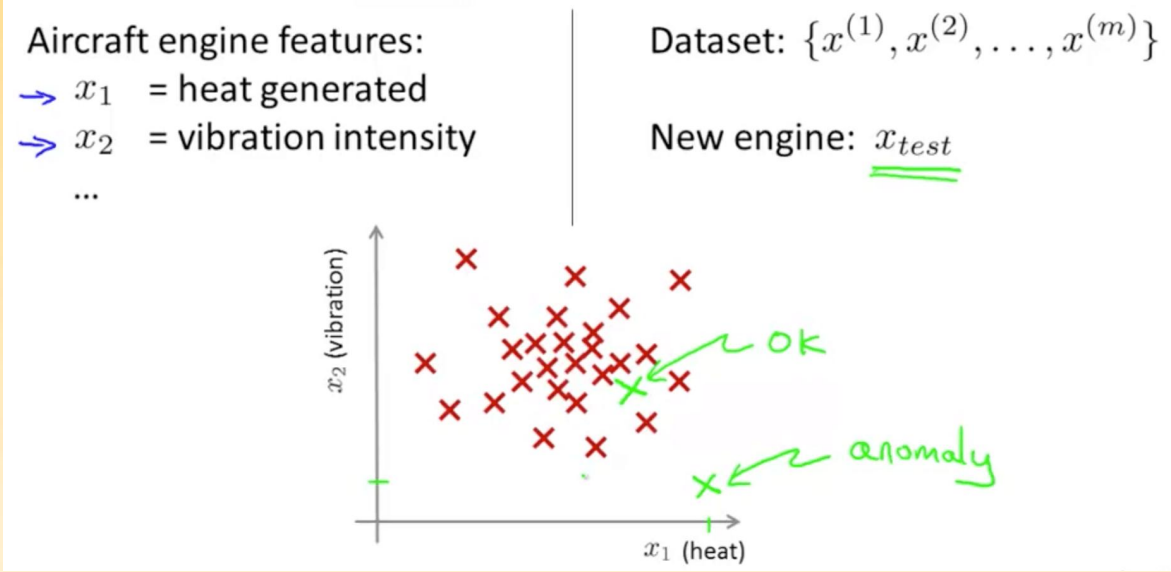




Unsupervised Learning

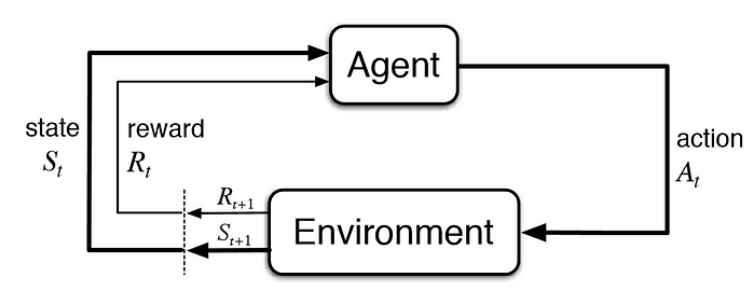
- **Unsupervised Learning** – learning is performed on unlabeled data in which the underlying structure is not previously known.
  - **Clustering** is an example of unsupervised learning where data is grouped based on its internal structure/features
  - **Anomaly Detection** analyzes some stream of input data to identify outliners in the data set
  - **Dimension reduction** used to narrow/clean data sets for tasks such as visualization or to improve the performance of other algorithms.

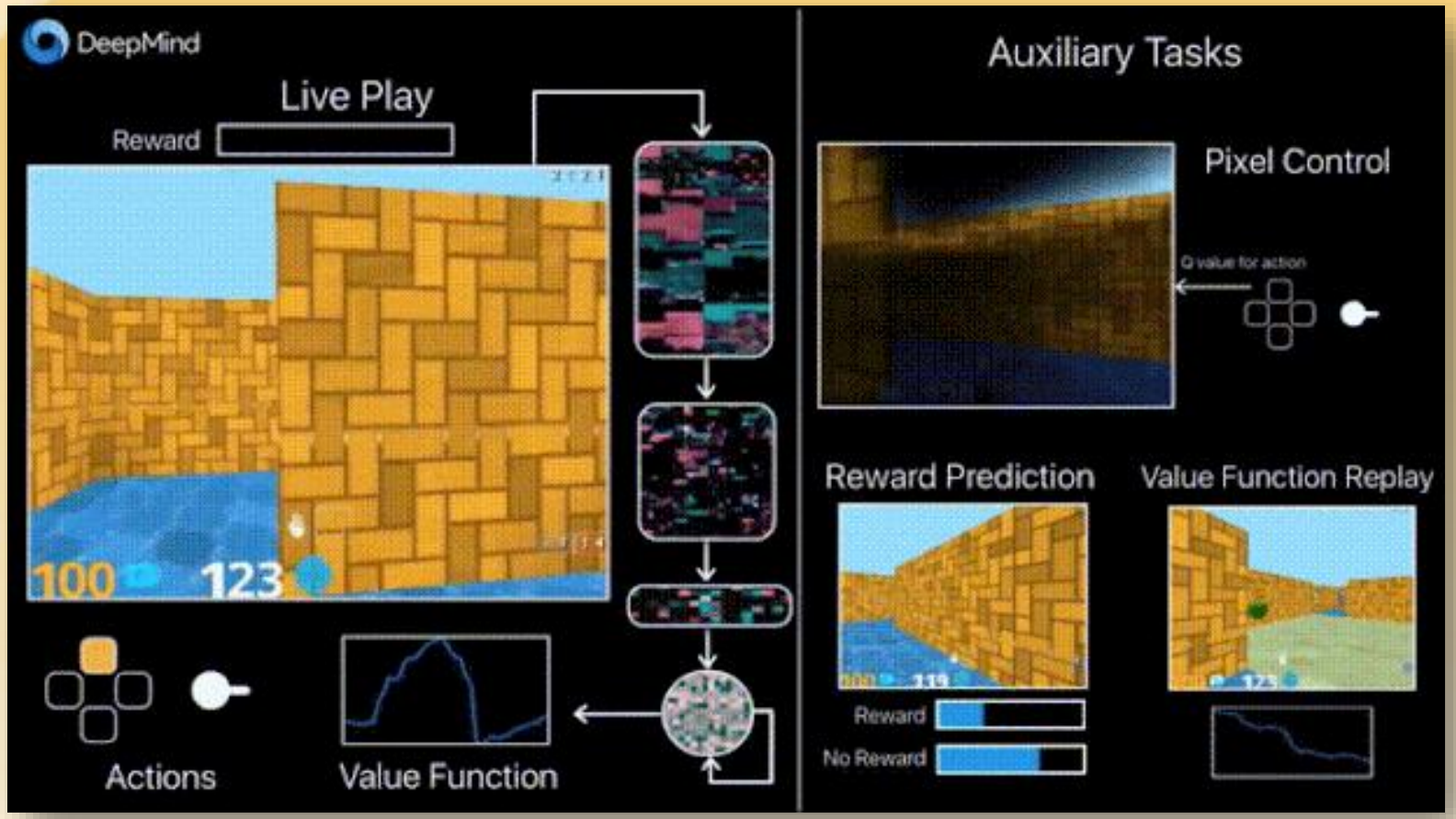




Reinforcement Learning

- **Reinforcement Learning** – learns to respond to its state and stimuli through the reinforcement of which actions provided favorable results vs. which options provided less favorable results.

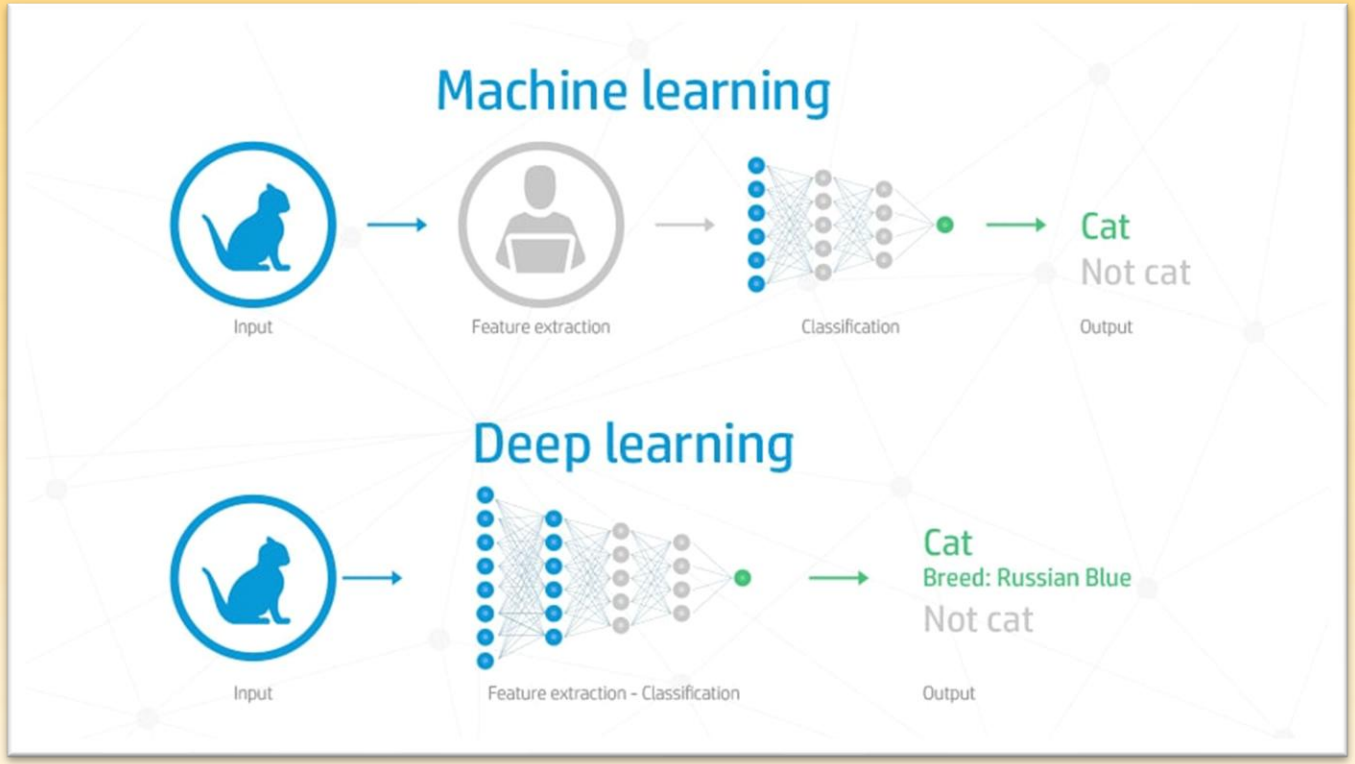




Deep Learning

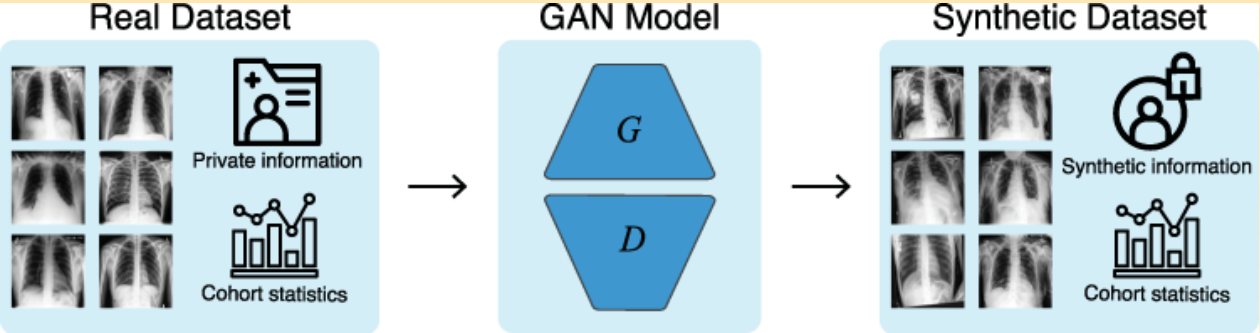
- Modern approach to ML that leverages "deep" neural networks to learn to **both** extract features from the source data and learn from the extracted features.
  - Traditional image-based classification vs. Deep image-based classification

- Applications include:
  - Image processing
  - Speech recognition
  - Natural Language Processing
  - Healthcare
  - Recommendations
  - Fraud detection



Generative AI

- **Generative AI** can generate text, images, video, and other media types by learning the characteristics of the training input and then producing an output with similar features
  - e.g. Generative Adversarial Networks (GANs) are a common example.
- Applications:
  - Synthetic data generation
  - Image correction
  - Deep fakes (for good and for bad)
  - AI art



Benefits of AI/ML

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Improved Data Analysis and Prediction

- Healthcare
- Finance

Improved Automation

- Manufacturing
- Customer Service

Personalization

- Retail
- Media Streaming
- Education

Adaptability

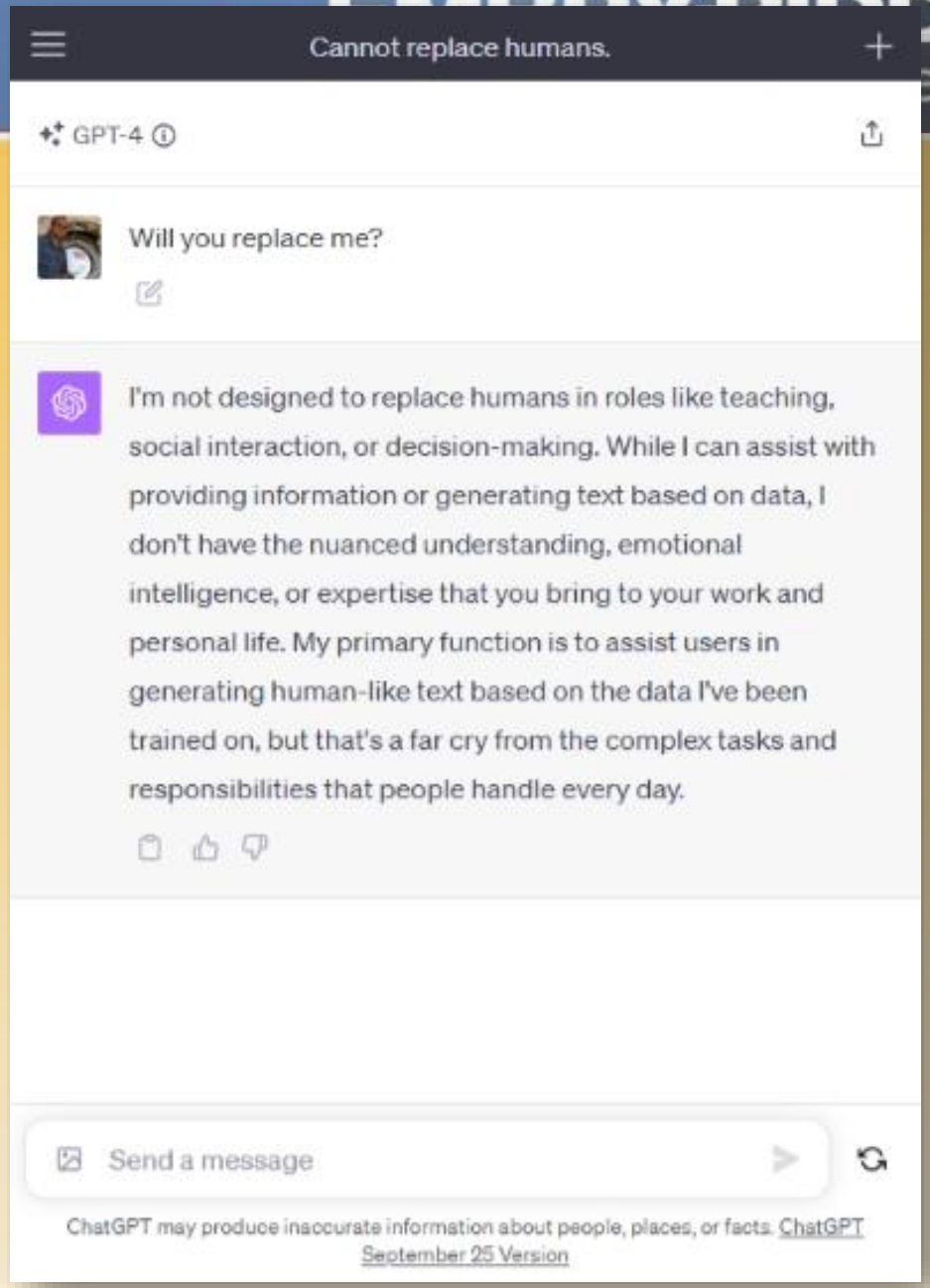
- Navigation Systems
- Natural Language Processing

Enhanced Decision Making

- Investment Planning
- Healthcare Diagnostics

Ethical Considerations

- Bias and Fairness
- Job Displacement
- Environmental Impact
- Consent and Data Ownership
- Transparency and Explainability
- Privacy
- Accountability and Responsibility
- Safety
- Global Collaboration and Regulation



AI/ML in Aviation and Aerospace

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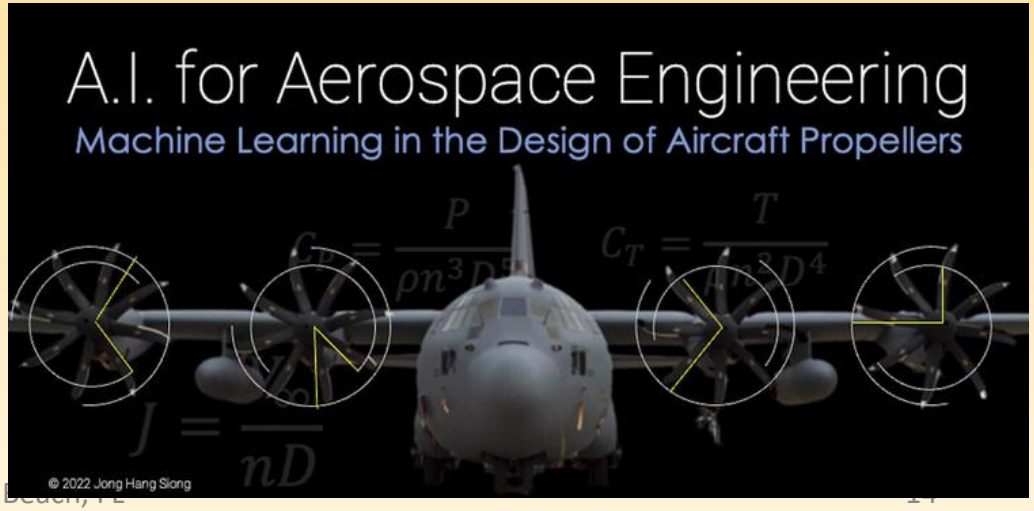

The Future of AI

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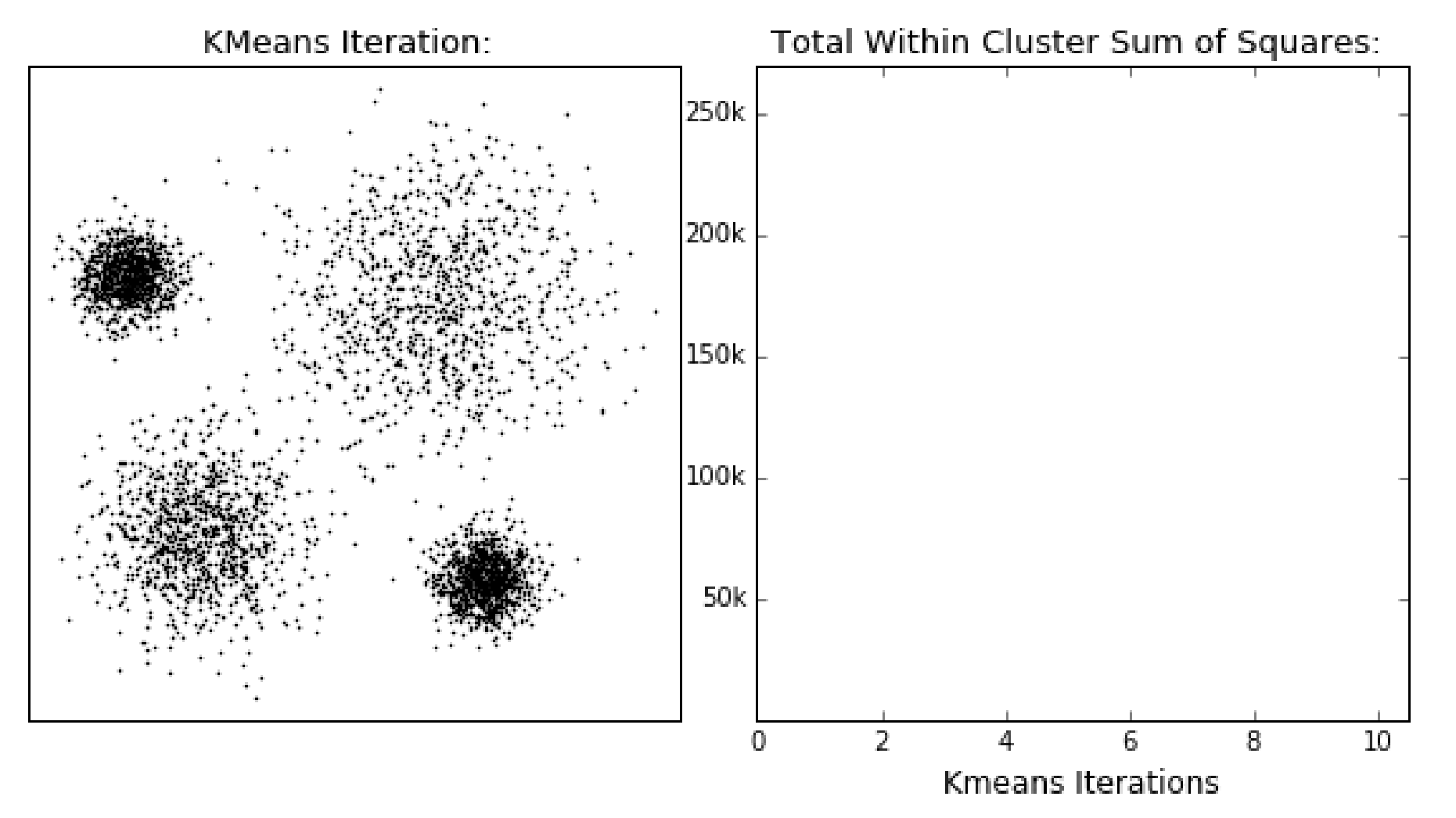
- Transportation
- Computing & Technology
- Finance
- Health Care
- Education
- Marketing
- Law Enforcement
- Academia
- Customer Service

- Impacts many sectors
- Examples of emerging use cases
  - AAM: Drone Delivery, Air-Taxi & Air-Metro Services
  - Medical imaging diagnostics
  - Tutoring and learning software (e.g., Duo Lingo)
  - Weather prediction/forecast
  - Automated design
  - Automated vehicles
  - Pursuit of AGI
  - Cybersecurity across all sectors

Clustering

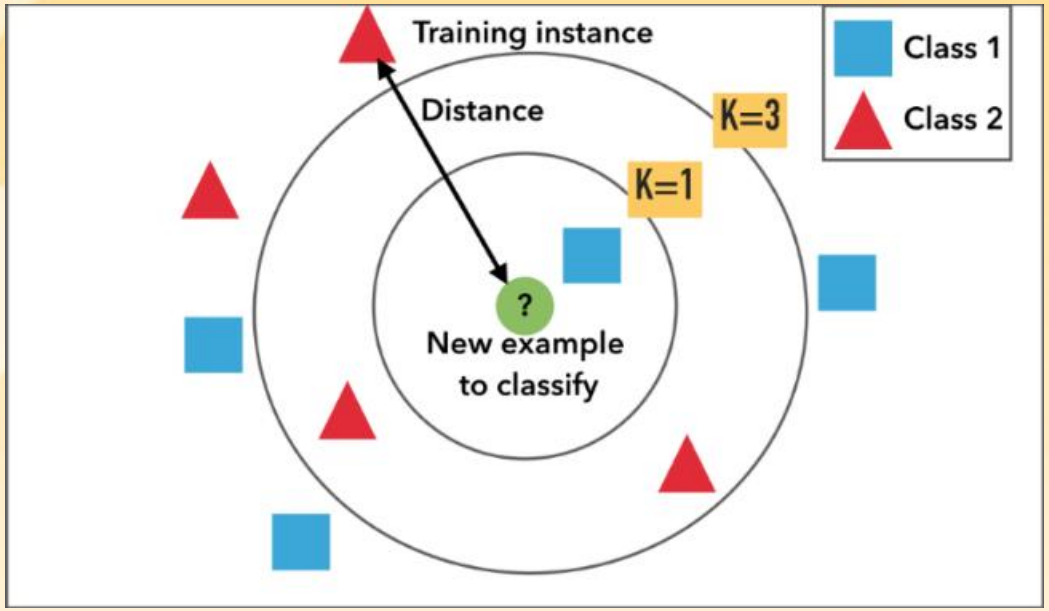

KMeans Iteration:
Total Within Cluster Sum of Squares:
250k
200k
150k
100k
50k
0
2
4
6
8
10
Kmeans Iterations

Reinforcement Learning

- Typically explained using agent paradigm
  - Agent must learn a policy, which dictates actions given state
  - Learning encourages maximization of rewards and/or minimization of penalties
- Both supervised and unsupervised RL techniques exist
- Reinforcement Learning Algorithms:
  - Monte Carlo
  - Q-Learning
  - Markov Decision Modeling (MDM)

Instance-based Algorithms

- Learn examples by heart
- Generalizes new data based upon some similarity metric
- Example
  - k-Nearest Neighbors utilizes the data and labels from past examples to identify the k neighbors of "smallest distance away" from a new input and use their labels to facilitate classification and/or prediction.

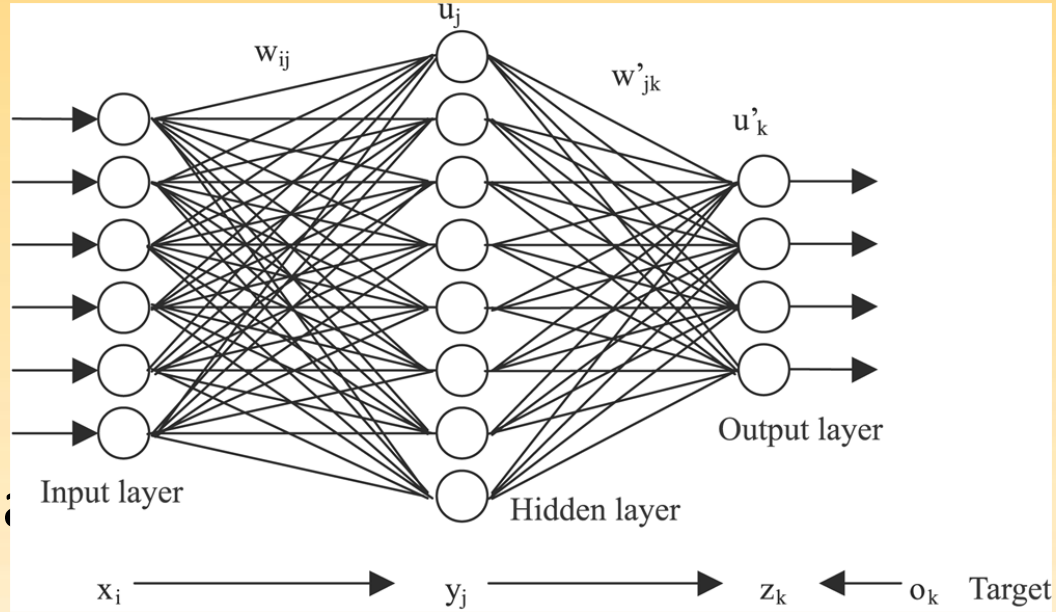


Model-based Learning

- From a set of examples, a model is constructed to make predictions based upon new input cases

- Model Elements:
  - Model selection
  - Model parameters
  - Utility or fitness function (positive evaluation)
    - Or, cost function (negative evaluation)

- Training is performed by feeding it data and a improve the model's fit to the data.



Pilot Program: ATCT AI @ Heathrow

- **AI System "Amy":** Tested at Heathrow Airport to assist air traffic controllers.
- **Purpose:** Manage UK's most congested airspace efficiently.
- **System Features**
  - Data Integration: Combines radar and video data for holistic aircraft visualization.
  - Digital Control Tower: Replaces traditional tower view with digital processing.
  - Real-Time Data: Provides flight number, aircraft type, and status (inbound/outbound).
- **Testing:** Over 40,000 flights tested.
- **Future Plans:** Fully operational digital contingency tower by 2027.
- **Limitations:** Lack of contextual judgment and data limitations.
- **Human Collaboration:** AI works alongside human controllers for better results.

London's Heathrow Airport trials AI technology to enhance air traffic control operations - YouTube

Air Traffic Management (ATM):

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- Innovations leading to more efficient, safer, and environmentally conscious aviation.
- Better management of **complex air traffic flow**, optimizing, sequencing and routing for efficiency and capacity.
- Adoption in **air traffic management**.
- Analysis of surveillance data to **detect conflicts and improve controller awareness** using neural networks.
- **Aviation weather** prediction

GE Aviation Predictive Maintenance

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- Two remote monitoring diagnostic centers provide round-the-clock monitoring of engine data.
  - Approx. 44,000 in-service engines monitored during flight
- Example 1: Anomaly Detection algorithm flags a fault oil sensor.
  - Alerts generated and dispatched to appropriate maintenance team(s)
  - Reduces lifecycle costs
    - Reduces high-cost repairs/replacement
    - Less downtime
- Example 2: AI-Enhanced Inspection
  - Inspection robots collect imagery of system elements
    - Optical imagery
    - X-ray fluorescence spectroscopy (XFS) on metal components
  - Imagery used to train models to detect part defects for improved future inspections

Source: GE Aerospace Expands Scope of Artificial Intelligence Support | Aviation International News

Predictive Maintenance

- Improvement of aviation services and guarantee of uniformity, including aircraft maintenance.
  - Prediction of **aircraft maintenance needs**.
  - Analysis of sensor data and prediction of **equipment malfunctions** for proactive maintenance.
  - Anticipation of **potential failures** by monitoring aircraft system's performance and usage data.

Advanced Space: AI/ML for Space GNC

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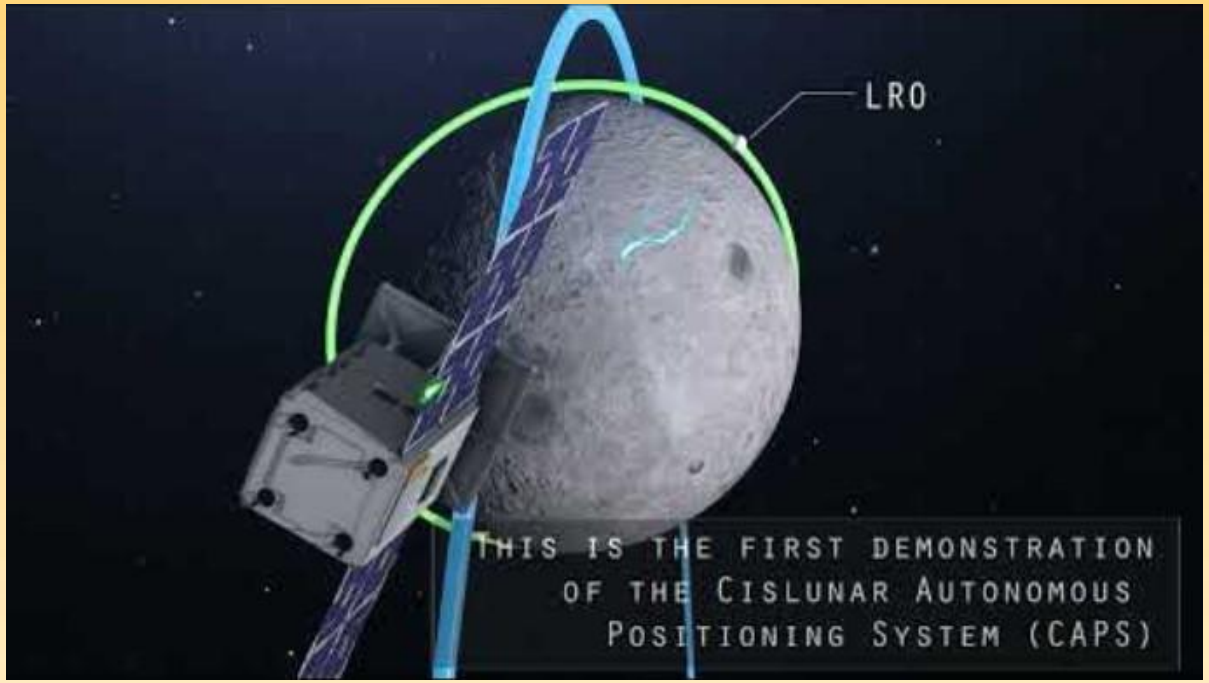
- Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft
- AI Systems
  - SigmaZero – anomaly detection and classification
    - e.g. identifying acceleration anomalies that might add error to course tracking.
    - Neural network driven
  - CAPS
    - Position system for cislunar travel in Earth-Moon system
    - Leverages Lunar Reconnaissance Orbiter and Deep Space Network from NASA
    - Measures gravity insatiability between Earth and moon.
  - NNEP
    - Enables optimal satellite maneuvering
    - Maneuver integrity with real-time monitoring
    - Electrical or chemical propulsion



Flight Control and Navigation:

- Analysis of sensor data, weather, and traffic for pilot assistance and route optimization.
- Enhancement of spaceship navigation and control by processing sensor data and orbital mechanics.
- Optimization of flight trajectories using machine learning.
- Enhancement of cockpit automation.

AI for CFD

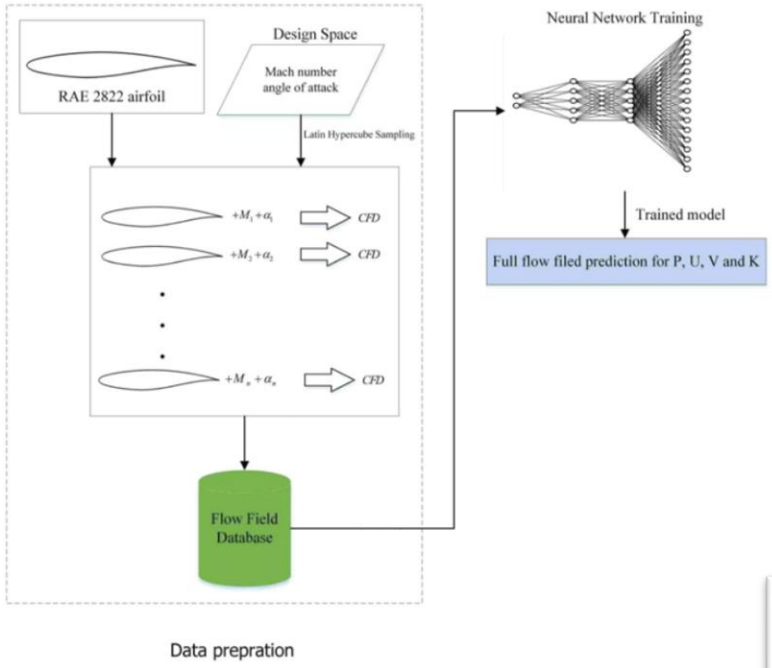


Fig. 2 Flowchart of machine learning based on direct CFD dataset

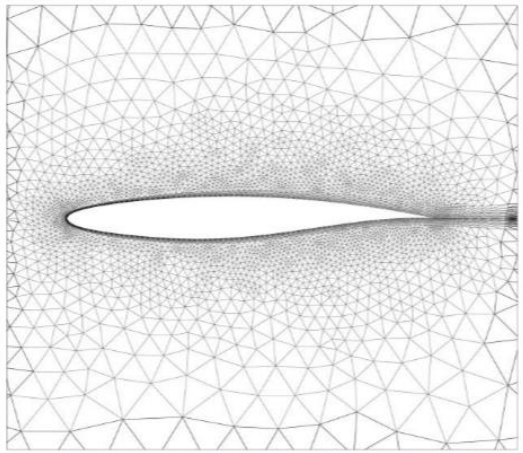


Fig. 4 Computational grid for RAE2822 used as neural network input

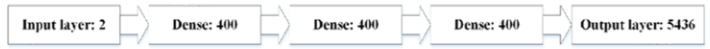


Fig. 5 MLP schematic structure

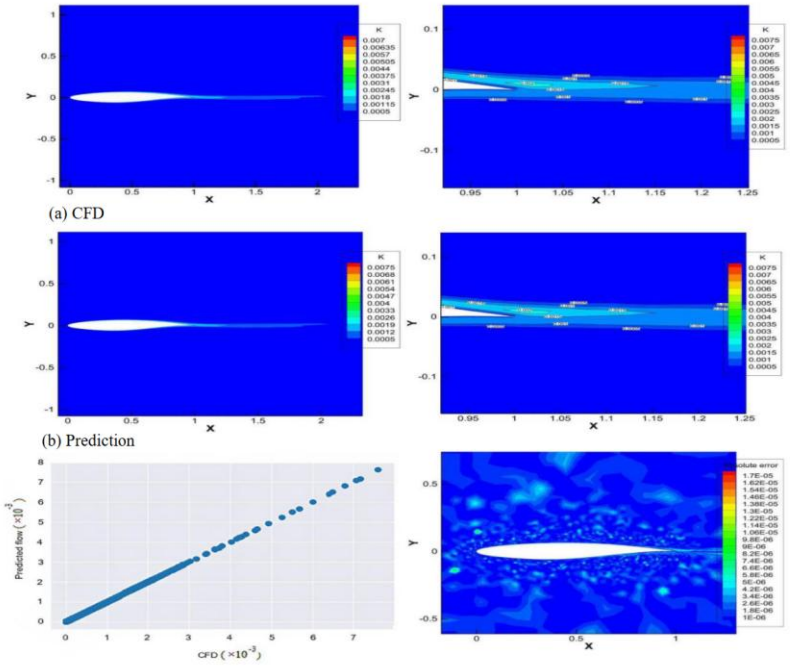


Fig. 7 Turbulence kinetic energy prediction for random testing case (M=0.45, AOA=4.00)

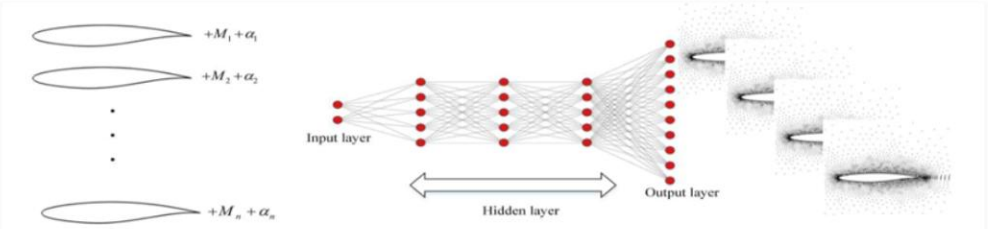


Table 3 Results for random testing cases

Flow field	Mach number	AOA	CFD time (s)	Prediction time	MSE (10^{-6})	Speed up
P	0.36	3.62	698	3.00	0.092	232
U	0.27	3.5	709	3.00	0.396	260
V	0.44	2.41	624	3.00	0.612	260
K	0.45	4.00	745	3.00	0.112	260

Nemati, M., & Jahangirian, A. (2023).

Aircraft Design and Optimization:

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- Acceleration of the design and testing of aircraft and spacecraft by evaluating simulations and optimizing aerodynamics.
- Design of more efficient aircraft.
- Enhancement of aerodynamics and aid in aircraft design.
- Structural and design optimization of aircraft components using machine learning.
- Application of deep learning in areas like aircraft design.
- Potential revolution in aircraft optimization for efficiency and aerodynamics.

Fuel Efficiency Optimization:

Passenger Experience and Customer Service

- Personalized Recommendations
- Customer Service Automation
- Behavior Analysis
- Improved airport operations to reduce arrival and departure delays

Aerospace Manufacturing

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- Automation of manufacturing.
- Application in smart manufacturing within the aerospace industry.
  - IoT components to monitor elements of manufacturing process, detecting faults in automated manufacturing systems and alert appropriate personnel.
- Simplification of previously intractable issues in areas like production using deep learning.
- Big data enables optimization in tool life cycles, supply chain management, production forecasting, throughput, etc.

Cybersecurity

- Enhancement of detection, prevention, and response to aerospace security risks, including aircraft asset protection.
- Monitoring of aviation and spacecraft systems for hostile behavior.
- Enhancement of cybersecurity protocols by identifying and reducing possible risks in aircraft communication systems.

Aviation Safety and Accident Investigation

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- Analyze flight recorder data to identify safety risks and aid in accident investigations.
- Examine accident data to foresee safety concerns and assist in creating safety protocols and emergency plans.
- Implement natural language processing (NLP) for aviation safety.
- Improve airport and aircraft security through equipment health monitoring.
- Leverage machine learning and NLP to anticipate human factors in aviation incident reports.
- Use NLP and machine learning to categorize human factor issues in incident reports.
- Identify anomalies in go-arounds by recognizing patterns with machine learning.
- Extract insights from aviation safety narratives using NLP and clustering techniques.
- Evaluate risk factors during flights using the SAFE methodology and machine learning.

UAV/Drone Applications

- Integration in UAVs and ATM systems significantly influencing unmanned aircraft.
- Enhancement of the performance of autonomous UAV networks.
- Facilitation of aircraft inspections using AI-driven UAVs.
- Automation of drone inspections, exams, and analysis of flight data.
- Automation of aircraft maintenance visual inspections using autonomous drones.

Pilot Training

- Optimization of flight paths and simulation of conditions for pilot training.
- Assistance with real-time decision-making during flights.
- Personalization of learning, provision of VR training, identification of weaknesses, and tracking of progress in aviation education.
- Offering of objective performance evaluation and personalized feedback in simulator-based pilot training using machine learning frameworks.
- Prediction of flight skills.

Thank you, any questions?

Backup Slides

Other Use Cases as Identified by Google Gemini

Warning – Gemini doesn't have a good filter, outputs in random order, and some non-sense errors coming after this point.

Why include this? I hope this provides some ideas that might spare your future goals, ambitions, or studies in the AI and Aviation world.

Other Applications:

- AI is used in the design and analysis of digital multiphase material microstructures.
- AI contributes to automation in executing flight routes.
- AI is being explored for collision avoidance in air traffic.
- AI can be used for predicting aircraft moving speed using inertial navigation.
- AI algorithms analyze reliability of predictive algorithms in aerospace.

Other Applications (continued)

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- AI supports digitalization in aerospace composites.
- AI is explored for remote assistance in industrial purposes.
- AI is used in man-machine air combat systems.
- AI can be applied for bus voltage compensation and power-sharing coefficient designs.
- AI aids the development of computing platforms using blockchain in aerospace.
- AI is incorporated for DoS detection in UAV systems.
- AI assists in downlink interference control in dense unmanned networks.

Other Applications (continued)

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- Machine learning can predict when computation offloading would be helpful for deep learning in networks.
- AI models can help in cooperatively getting context data in vehicular communication networks.
- AI can guide the development of data for thermal insulators.
- Deep neural networks are used for design space exploration for DNN inference.
- Machine learning algorithms can improve resource allocation.
- Extreme learning machines offer high-performance, low-power hardware for machine learning.

Other Applications (continued)

- AI-based methods are used to reduce the number of robots required for a performance level in multi-robot systems.
- Ant colony optimization can reduce the number of planes needed by calculating efficient flight paths.
- IoT-based data-driven models offer PHM solutions for aircraft maintenance.
- NLP is used to predict failure of infrastructure.
- AI can support operational-level decision-making for aviation routes.

Other Applications (continued)

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- Recurrent neural networks categorize error patterns and monitor air traffic.
- AI can automate ATC systems for safe aircraft grouping and sequencing.
- Machine learning enables prediction of failures hundreds of seconds in advance.
- Computer vision and deep learning improve visual bridge inspection accuracy for UAVs.
- AI can be used in water management process systems via remote sensing.
- Environmental drones with AI can do spot checks on air quality.

Other Applications (continued)

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- Metaheuristic algorithms aid in aircraft maintenance time and cost optimization.
- Explainable reinforcement learning contributes to fully automated and dependable autonomous aircraft.
- Particle swarm optimization optimizes fleet assignment, reducing daily costs.
- Neural networks power morphing unmanned aerial vehicles.
- Machine learning algorithms using the shortest path method help airlines save money on fuel, time, and seat capacity.

Other Applications (continued)

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- Recurrent Neural Networks (LSTM) predict turboprop engine exhaust emissions and combustion efficiency.
- AI is transforming the aircraft sector by enhancing various operations, including maintenance, air traffic control, and customer experience.
- AI-driven threat detection systems assess data from various sensors for enhanced security.
- AI can operate and manage autonomous defense systems to neutralize aerial threats.
- AI offers decision assistance to security professionals in high-pressure scenarios.
- AI integrates data from multiple sources to provide a comprehensive perspective on potential dangers.

Other Applications (continued)

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- AI optimizes launch trajectories, improves spacecraft health monitoring, and customizes experiences for space travelers.
- AI-powered robots can perform tasks like satellite repairs and asteroid maneuvering with high accuracy.
- AI-powered training programs revolutionize preparation for space tourists and explorers using VR and AR.
- AI is expected to foster innovation and breakthroughs in space travel and exploration.
- AI advancements can help hypersonic aircraft travel vast distances rapidly.

Other Applications (continued)

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- Data mining methods on big data can optimize in-flight costs for air traffic flow management.
- AI can aid beyond visual range target allocation and multi-aircraft collaborative decision-making.
- Self-driving aircraft towing vehicles are being explored.
- Data-driven methods using probabilistic clustering and dimensionality reduction can monitor satellite housekeeping data.
- AI is being used to evaluate and analyze augmented reality for remote help in the workplace, reducing production stoppages.

Other Applications (continued)

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- Deep Q-Networks are used for controlling aircraft landing.
- Reinforcement learning has demonstrated capability in man-machine air combat systems.
- AI design approaches are suggested for optimal bus voltage compensation and power sharing.
- AI can be combined with ACARS signals for enhanced data transmission.
- Blockchain technology in aerospace is being explored with AI aiding its development.

Other Applications (continued)

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- Machine learning techniques are used for IoT intrusion detection.
- SVM is a classifier used for minimized classification risk.
- Feature extraction is a key step in data pre-processing for machine learning.
- Various features are extracted from network data for intrusion detection.
- AI competency frameworks are being developed for aviation education.
- Current aviation bachelor's programs may not sufficiently develop needed AI competencies.

Other Applications (continued)

EMBRY-RIDDLE Aeronautical University

- Dimensionality reduction simplifies variables while capturing essential variations.
- Tensor decomposition is a dimension reduction technique.
- Expert participation and verbal scales are important in evaluating complex technical objects with uncertainty.
- Interval type-II fuzzy sets help in aggregating individual expert information for a generalized model.
- Methods for formalizing group expert information based on interval type-II fuzzy sets have been developed.
- Regression models based on interval type-II fuzzy sets are used for predicting expert evaluations.

Other Applications (continued)

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- Aviation educators need to make evidence-based changes to curricula based on AI integration.
- Competency frameworks for AI in aviation are developed using inductive analysis.
- Analyzing existing aviation curricula is a key step in integrating AI education.
- Integrating AI in aviation requires building critical sets of competencies.
- Text analysis tools can visualize word frequencies and co-occurrences in aviation engineering corpora.
- Learning outcomes should demonstrate both conceptual knowledge and practical application of AI in aviation.

Other Applications (continued)

EMBRY-RIDDLE Aeronautical University

- The annotation of AI/ML key terms can involve subjectivity due to natural language ambiguity.
- PCA assumes linearity and may overestimate dimensions in non-linear data.
- Non-linear dimensionality reduction methods like Kernel PCA can address PCA limitations.
- A structured literature review, analysis of aviation authority documents, and expert interviews are used to identify AI competency needs.
- Keywords like "Aerospace Engineering," "Artificial Intelligence," and related terms are used to identify relevant research.
- Challenges in aerospace include optimizing competing goals and managing dynamic, unpredictable air traffic situations.

Other Applications (continued)

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- AI can enhance automation, pilot-AI interaction, and situational awareness.
- Limited understanding of aviation-specific jargon can hinder AI in communication analysis.
- AI needs to handle real-time dynamic scenarios like sudden weather disasters.
- The adoption of aerospace AI necessitates investments in hardware, software, and data, along with a data governance framework.
- A culture fostering creativity and cooperation is needed for aerospace AI adoption.

Other Applications (continued)

EMBRY-RIDDLE Aeronautical University

- Collaboration with research institutes and academic experts can help stay updated with AI advancements.
- A comprehensive approach integrating AI, process innovation, and other factors enhances global competitiveness for aerospace companies.
- Proper data governance and quality are essential for the dependability and trustworthiness of AI systems in aerospace.
- International cooperation can speed up development in AI safety and security for aerospace.
- AI presents diverse functionalities that can significantly enhance defense mechanisms against various threats.
- The convergence of factors is leading to innovative concepts in space tourism.
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- Design and analysis of digital multiphase material microstructures.
- Automation in executing flight routes.
- Exploration for collision avoidance in air traffic.
- Prediction of aircraft moving speed using inertial navigation.
- Analysis of reliability of predictive algorithms in aerospace.
- Support for digitalization in aerospace composites.
- Exploration for remote assistance in industrial purposes.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Application in man-machine air combat systems.
- Application for bus voltage compensation and power-sharing coefficient designs.
- Aid in the development of computing platforms using blockchain in aerospace.
- Incorporation for DoS detection in UAV systems.
- Assistance in downlink interference control in dense unmanned networks.

Other Applications: Continued

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- Prediction of when computation offloading would be helpful for deep learning in networks using machine learning.
- Cooperative context data acquisition in vehicular communication networks.
- Guidance in the development of data for thermal insulators.
- Design space exploration for DNN inference using deep neural networks.
- Improvement of resource allocation using machine learning algorithms.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Offering of high-performance, low-power hardware for machine learning using extreme learning machines.
- Reduction of the number of robots required for a performance level in multi-robot systems using AI-based methods.
- Reduction of the number of planes needed by calculating efficient flight paths using ant colony optimization.
- Offering of PHM solutions for aircraft maintenance using IoT-based data-driven models.
- Prediction of failure of infrastructure using NLP.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Support for operational-level decision-making for aviation routes.
- Categorization of error patterns and monitoring of air traffic using recurrent neural networks.
- Automation of ATC systems for safe aircraft grouping and sequencing.
- Prediction of failures hundreds of seconds in advance using machine learning.
- Improvement of visual bridge inspection accuracy for UAVs using computer vision and deep learning.

Other Applications: Continued

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- Application in water management process systems via remote sensing.
- Spot checks on air quality using environmental drones with AI.
- Use for artificial air monitoring.
- Aid in aircraft maintenance time and cost optimization using metaheuristic algorithms.
- Contribution to fully automated and dependable autonomous aircraft using explainable reinforcement learning.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Optimization of fleet assignment, reducing daily costs using particle swarm optimization.
- Powering of morphing unmanned aerial vehicles using neural networks.
- Saving money on fuel, time, and seat capacity using machine learning algorithms with the shortest path method.
- Prediction of turboprop engine exhaust emissions and combustion efficiency using recurrent neural networks (LSTM).

Other Applications: Continued

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- Transformation of the aircraft sector by enhancing various operations, including maintenance, air traffic control, and customer experience.
- Assessment of data from various sensors for enhanced security using AI-driven threat detection systems.
- Operation and management of autonomous defense systems to neutralize aerial threats.
- Offering of decision assistance to security professionals in high-pressure scenarios.
- Integration of data from multiple sources to provide a comprehensive perspective on potential dangers.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Optimization of launch trajectories, improvement of spacecraft health monitoring, and customization of experiences for space travelers.
- Performance of tasks like satellite repairs and asteroid maneuvering with high accuracy using AI-powered robots.
- Revolutionization of preparation for space tourists and explorers using VR and AR with AI-powered training programs.
- Fostering of innovation and breakthroughs in space travel and exploration.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Assistance in hypersonic aircraft travel over vast distances rapidly.
- Optimization of in-flight costs for air traffic flow management using data mining methods on big data.
- Aid in beyond visual range target allocation and multi-aircraft collaborative decision-making.
- Exploration of self-driving aircraft towing vehicles.
- Monitoring of satellite housekeeping data using data-driven methods with probabilistic clustering and dimensionality reduction.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Evaluation and analysis of augmented reality for remote help in the workplace, reducing production stoppages.
- Control of aircraft landing using deep Q-networks.
- Demonstration of capability in man-machine air combat systems using reinforcement learning.
- Suggestion of design approaches for optimal bus voltage compensation and power sharing.
- Combination with ACARS signals for enhanced data transmission.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Exploration of blockchain technology in aerospace with AI aiding its development.
- Use of machine learning techniques for IoT intrusion detection.
- Classification using SVM for minimized classification risk.
- Extraction of features from network data for intrusion detection.
- Development of AI competency frameworks for aviation education.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Insufficient development of needed AI competencies in current aviation bachelor's programs.
- Simplification of variables while capturing essential variations using dimensionality reduction.
- Use of tensor decomposition as a dimension reduction technique.
- Importance of expert participation and verbal scales in evaluating complex technical objects with uncertainty.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Aggregation of individual expert information for a generalized model using interval type-II fuzzy sets.
- Formalization of group expert information based on interval type-II fuzzy sets.
- Prediction of expert evaluations using regression models based on interval type-II fuzzy sets.
- Need for aviation educators to make evidence-based changes to curricula based on AI integration.
- Development of competency frameworks for AI in aviation using inductive analysis.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Analysis of existing aviation curricula as a key step in integrating AI education.
- Building of critical sets of competencies for integrating AI in aviation.
- Visualization of word frequencies and co-occurrences in aviation engineering corpora using text analysis tools.
- Demonstration of both conceptual knowledge and practical application of AI in aviation through learning outcomes.
- Subjectivity in the annotation of AI/ML key terms due to natural language ambiguity.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Overestimation of dimensions in non-linear data using PCA.
- Addressing of PCA limitations using non-linear dimensionality reduction methods like Kernel PCA.
- Identification of AI competency needs using a structured literature review, analysis of aviation authority documents, and expert interviews.
- Identification of relevant research using keywords like "Aerospace Engineering," "Artificial Intelligence," and related terms.
- Optimization of competing goals and management of dynamic, unpredictable air traffic situations as challenges in aerospace.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Enhancement of automation, pilot-AI interaction, and situational awareness.
- Hindrance in communication analysis due to limited understanding of aviation-specific jargon.
- Handling of real-time dynamic scenarios like sudden weather disasters.
- Necessity of investments in hardware, software, and data, along with a data governance framework for the adoption of aerospace AI.
- Need for a culture fostering creativity and cooperation for aerospace AI adoption.
- Collaboration with research institutes and academic experts to stay updated with AI advancements.

Other Applications: Continued

EMBRY-RIDDLE Aeronautical University

- Enhancement of global competitiveness for aerospace companies through a comprehensive approach integrating AI, process innovation, and other factors.
- Speeding up of development in AI safety and security for aerospace through international cooperation.
- Enhancement of defense mechanisms against various threats using diverse AI functionalities.